

## Assingment -8

1. Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
1 row in set (0.01 sec)

KD3-98829-upendra> select * from orders where cnum = (select cnum from customers where cname = 'Cisneros');
+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+
| 3001 | 18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)

KD3-98829-upendra>
```

7. Extract all the orders of Motika.

```
KD3-98829-upendra> select * from orders where cnum = (select snum from salespeople where sname = 'Motika');
Empty set (0.00 sec)

KD3-98829-upendra> select * from orders where snum = (select snum from salespeople where sname = 'Motika');
+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

KD3-98829-upendra>
```

8. Find all the orders of salespeople servicing customers in London.

```
the right syntax to use near '(select distinct snum from customers
KD3-98829-upendra> select * from orders where snum = (select distinct snum from salespeople where sname = 'Motika');
+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

KD3-98829-upendra>
```

6. Write a query that selects all orders for amounts greater than any for the customers in London.

```

3 rows in set (0.00 sec)

KD3-98829-upendra> select * from orders where amt > ANY(select amt from orders where cnum IN ( select cnum from customers where city ='London'));
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '' at line 1
KD3-98829-upendra> select * from orders where amt > ANY(select amt from orders where cnum IN ( select cnum from customers where city ='London'));
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+-----+-----+-----+
6 rows in set (0.01 sec)

KD3-98829-upendra>

```

4. Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'. (Hint: find all customers of serres salesperson).

```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '' at line 1
KD3-98829-upendra> select * from customer where rating IN(select rating from customers where snum =(select snum from salespeople where sname = 'Serres'));
ERROR 1146 (42502): Table 'sales.customer' doesn't exist
KD3-98829-upendra> select * from customers where rating IN(select rating from customers where snum =(select snum from salespeople where sname = 'Serres'));
+-----+-----+-----+-----+-----+
| cnum | cname   | city   | rating | snum |
+-----+-----+-----+-----+
| 2002 | Giovanni | Rome   | 200    | 1003 |
| 2003 | Liu      | San Jose | 200    | 1002 |
| 2004 | Grass    | Berlin | 300    | 1002 |
| 2008 | Cisneros | San Jose | 300    | 1007 |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)

KD3-98829-upendra>

```

11. Select customers who have a greater rating than any other customer in Rome.

```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'IN (select distinct rating from customers where city ='Rome')' at line 1
KD3-98829-upendra> select * from customers where rating >ANY (select distinct rating from customers where city ='Rome');
+-----+-----+-----+-----+-----+
| cnum | cname   | city   | rating | snum |
+-----+-----+-----+-----+
| 2002 | Giovanni | Rome   | 200    | 1003 |
| 2003 | Liu      | San Jose | 200    | 1002 |
| 2004 | Grass    | Berlin | 300    | 1002 |
| 2008 | Cisneros | San Jose | 300    | 1007 |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)

KD3-98829-upendra>

```

12. Select all orders that had amounts that were greater than at least one of the orders from '1990-10-04'.

```
KD3-98829-upendra> select * from orders where amt > ANY ( select amt from orders where odate = '1990-10-04');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for
the right syntax to use near 'ANY ( select amt from orders where odate = '1990-10-04')' at line 1
KD3-98829-upendra> select * from orders where amt > ANY ( select amt from orders where odate = '1990-10-04');
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 | 309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+-----+-----+-----+
8 rows in set (0.00 sec)

KD3-98829-upendra>
```

13. Find all orders with amounts smaller than any amount for a customer in San Jose.

```
KD3-98829-upendra> select amt from orders where snum = (select cnum from customers where city = 'San Jose');
ERROR 1242 (21000): Subquery returns more than 1 row
KD3-98829-upendra> select amt from orders where cnum IN( select cnum from customers where city = 'San Jose');
+-----+
| amt |
+-----+
| 18.69 |
| 5160.45 |
| 1098.16 |
+-----+
3 rows in set (0.00 sec)

KD3-98829-upendra> select * from orders where amt < ANY( select amt from orders where cnum IN( select cnum from customers where city = 'San Jose'));
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+
| 3001 | 18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 | 75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 | 309.95 | 1990-10-04 | 2004 | 1002 |
+-----+-----+-----+-----+
8 rows in set (0.00 sec)

KD3-98829-upendra>
```

2. Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table

```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that
(select avg(amt) from or' at line 2
KD3-98829-upendra> SELECT o.amt, c.cname, c.rating
-> FROM customers c
-> INNER JOIN orders o ON c.cnum = o.cnum
-> WHERE o.amt > (SELECT AVG(amt) FROM orders);
+-----+-----+-----+
| amt      | cname  | rating |
+-----+-----+-----+
| 5160.45  | Liu    | 200    |
| 9891.88  | Clemens| 100    |
| 4723.00  | Clemens| 100    |
+-----+-----+-----+
3 rows in set (0.00 sec)

KD3-98829-upendra> 

```

14. Select those customers whose rating are higher than every customer in London.

```

Empty set (0.00 sec)

KD3-98829-upendra> select * from customers where rating > any (select rating from custo
+-----+-----+-----+-----+-----+
| cnum | cname   | city    | rating | snum |
+-----+-----+-----+-----+-----+
| 2002 | Giovanni| Rome    | 200    | 1003 |
| 2003 | Liu     | San Jose| 200    | 1002 |
| 2004 | Grass   | Berlin  | 300    | 1002 |
| 2008 | Cisneros| San Jose| 300    | 1007 |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

KD3-98829-upendra> 

```

5. Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).

```

2 rows in set (0.00 sec)

KD3-98829-upendra> select * from salespeople s where s.snum not in (Select c.snum from customers c wh
re c.city = s.city);
+-----+-----+-----+-----+
| snum | sname   | city    | comm |
+-----+-----+-----+-----+
| 1004 | Motika  | London  | 0.11 |
| 1007 | Rifkin  | Barcelona| 0.15 |
| 1003 | Axelrod | New York | 0.10 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

KD3-98829-upendra> 

```

9. Find names and numbers of all salesperson who have more than one customer.

```
2 rows in set (0.00 sec)

KD3-98829-upendra> select sname ,snum from salespeople where snum IN (select snum from customers group
p by snum having count(cname)>1);
+-----+-----+
| sname | snum |
+-----+-----+
| Peel  | 1001 |
| Serres| 1002 |
+-----+-----+
2 rows in set (0.01 sec)

KD3-98829-upendra>
```

10. Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```
1002 |
-----+
2 rows in set (0.00 sec)

D3-98829-upendra> select snum , sname, city from salespeople where snum IN(
->
-> select snum from customers
-> group by snum having count( cname)>1);
+-----+-----+-----+
| snum | sname | city |
+-----+-----+-----+
| 1001 | Peel  | London |
| 1002 | Serres| San Jose |
+-----+-----+-----+
2 rows in set (0.00 sec)

D3-98829-upendra>
```